

SOFTWARE REQUIREMENTS SPECIFICATION

RFP Analyzer — AI-Powered RFP Qualification & Compliance Screening Tool

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Project	SPS RFP Analyzer (Qualification Engine)
Parent initiative	SPS Proposal Capture System (8-stage assignment)
Document scope	Covers Stages 1–3 of the parent system: RFP Ingestion, Qualification Engine, and the Compliance Checklist portion of Proposal Planning
Status	Implemented and tested; Stages 4–7 (Knowledge Retrieval, Proposal Generation, Review/Refinement, Orchestration) are out of scope for this document — see Section 2.5

1. Introduction

1.1 Purpose

This document specifies the software requirements for the RFP Analyzer, a tool that reads an incoming Request for Proposal (RFP) document and produces a structured qualification assessment: a fit score, a go/no-go recommendation, a department-by-department compliance checklist, and supporting detail (deliverables, evaluation criteria, key dates, risks, and strengths). It is intended for a Proposal Capture Manager deciding whether SPS should pursue a given bid.

1.2 Scope

The RFP Analyzer is one component of a larger 8-stage “SPS Proposal Capture System” assignment. This SRS covers only the portion that has been designed, implemented, and tested to date:

- RFP Ingestion — extracting text from an uploaded RFP PDF.
- Qualification Engine — producing a fit score, GO/CONDITIONAL/NO-GO verdict, deliverables, evaluation criteria, key dates/budget, strengths, and risks.
- Compliance Checklist (part of Proposal Planning) — evaluating the RFP against SPS's fixed 35-item checklist, department by department, with evidence and page references.

Proposal Generation, Knowledge Retrieval, Review/Refinement, and Orchestration (Stages 4–7 of the parent assignment) are explicitly out of scope for this document; see Section 2.5 for how this system is expected to extend into them.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
RFP	Request for Proposal — a procurement document a client issues inviting vendors to bid.
SPS	The company on whose behalf this tool screens incoming RFPs (identity & access management / cybersecurity services provider).
Fit Score	A 0–100 score representing overall suitability of the RFP for SPS to bid on.
Verdict	The overall recommendation: GO, CONDITIONAL, or NO-GO.
Checklist Item	One line item from SPS's fixed 35-item Financial/Legal/Operations/Technical checklist.
Item Status	Per-item result: GO (requirement met), NO-GO (requirement failed / threshold exceeded), or REVIEW (needs human judgment or RFP is silent on it).
Company Profile	User-editable data describing SPS's capabilities, thresholds, and certifications, used to judge fit against a given RFP.
LLM	Large Language Model — here, Google's Gemini 2.5 Flash.

1.4 References

- SPS Financial/Legal/Operations/Technical RFP Checklist (source PDF, ported into checklist_items.py).
- Google Gemini API documentation (google-genai SDK).

- Streamlit documentation.

1.5 Overview

Section 2 describes the product at a high level: its context, functions, users, constraints, and assumptions. Section 3 specifies detailed functional and non-functional requirements. Section 4 describes the system architecture. Section 5 covers data requirements, including the full 35-item checklist. Section 6 lists known limitations.

2. Overall Description

2.1 Product Perspective

The RFP Analyzer is a standalone Python/Streamlit web application. It has no dependency on other SPS systems in its current form (Stage 1–3 scope) — it runs locally or on a single server, accepts a manually uploaded PDF, and calls the Gemini API directly. It is designed to later become one stage within a larger orchestrated pipeline (see Section 2.5), but does not depend on that pipeline existing to function today.

2.2 Product Functions

At a high level, the system:

1. Accepts an RFP PDF upload (or a bundled sample RFP for testing).
2. Extracts the RFP's text, preserving page boundaries.
3. Accepts an editable Company Profile (services, experience, insurance capacity, payment-term tolerance, certifications, revenue).
4. Sends the RFP text and Company Profile to Gemini in five separate calls: one for the overall qualification assessment, and four more — one per department — for the compliance checklist.
5. Applies two deterministic, code-level threshold rules (payment terms, insurance) that override the model's judgment on those two specific items.
6. Computes department-level and overall compliance scores directly from the checklist results (not separately judged by the AI).
7. Displays results in a tabbed interface (Deliverables, Evaluation Criteria, Compliance Checklist, Dates & Budget, Strengths & Risks).
8. Exports the completed analysis as a Markdown file or a styled PDF report.

2.3 User Characteristics

The primary user is a Proposal Capture Manager or similar role at SPS — someone who needs to quickly decide whether to pursue an RFP and hand off compliance obligations to the right department. No software engineering knowledge is assumed; the interface is a standard web form (upload, edit fields, click Analyze, read results, download a report).

2.4 Constraints

- Depends on an external, third-party LLM API (Google Gemini) being available and within its rate/quota limits.
- Gemini's free tier is rate-limited (approximately 10 requests/minute, 500/day on gemini-2.5-flash as of mid-2026); heavier use requires a billed Google Cloud project.
- Gemini's structured-output (controlled generation) feature has an internal complexity ceiling: a single schema requiring an exact-length array of many complex nested objects can be rejected with a “too many states for serving” error. This is why the compliance checklist is split into four department-scoped calls rather than one 35-item call (see Section 4).
- PDF text extraction (pdfplumber) requires a text-based PDF; scanned/image-only PDFs are not supported without OCR, which is not currently implemented.
- The RFP checklist itself (35 items, 4 departments) is fixed in code (checklist_items.py) and is specific to SPS's current policies; a different set of checklist items requires a code change, not just a configuration change.

2.5 Assumptions and Dependencies

This system assumes it will eventually sit inside a larger pipeline (RFP Ingestion → Qualification → Planning → Knowledge Retrieval → Proposal Generation → Review → Orchestration), per the full SPS Proposal Capture System assignment. The output of this system (the qualification result and compliance checklist) is intended to become an input to a future Proposal Generation stage. That stage, and the Knowledge Retrieval, Review/Refinement, and Orchestration stages, are not yet built and are assumed to be separate future work, not covered by this SRS.

3. Specific Requirements

3.1 Functional Requirements

FR-1: RFP Ingestion

ID	Requirement
FR-1.1	The system shall accept an RFP document as a PDF file upload through the web interface.
FR-1.2	The system shall extract text from every page of the uploaded PDF, retaining page-boundary markers so later stages can cite page numbers.
FR-1.3	The system shall provide a bundled sample RFP that can be loaded with one click, for testing without needing a real document.
FR-1.4	If a PDF contains no extractable text (e.g. a scanned image), the system shall display a clear error rather than proceeding silently with empty input.

FR-2: Company Profile Configuration

ID	Requirement
FR-2.1	The system shall provide an editable Company Profile with, at minimum: company name, services/capabilities, years of relevant experience, maximum insurance coverage available, acceptable payment terms (days), certifications, and annual revenue.
FR-2.2	The system shall pre-populate the Company Profile with sensible SPS-specific defaults, editable per analysis run.
FR-2.3	The Company Profile's values shall be used both by the AI-driven qualification assessment and by the deterministic threshold rules (FR-4).

FR-3: Qualification Assessment

ID	Requirement
FR-3.1	The system shall produce a 0–100 fit score and an overall verdict of GO, CONDITIONAL, or NO-GO, with a plain-language justification.
FR-3.2	The system shall extract a list of deliverables from the RFP, each flagged mandatory or optional, with a best-effort effort estimate in weeks.
FR-3.3	The system shall extract the RFP's weighted evaluation criteria, ordered by weight where stated.
FR-3.4	The system shall extract key dates and budget figures: submission deadline, contract value, payment terms, insurance requirement, and bond requirement, where stated.
FR-3.5	The system shall produce 3–6 Strengths and 3–6 Risks/Weaknesses, each with a severity (for risks) and a supporting note.

FR-4: Compliance Checklist

ID	Requirement
FR-4.1	The system shall evaluate the RFP against a fixed, literal 35-item checklist spanning four departments: Financial/Accounting (6), Legal (13), Operations (11), Technical (5).
FR-4.2	Each checklist item's result shall include: an item status (GO / NO-GO / REVIEW), a plain-language reason, supporting evidence quoted or paraphrased from the RFP (or null if not addressed), and a page reference where determinable.
FR-4.3	The system shall always return exactly 35 checklist results, regardless of how the underlying model responds; any item the model fails to address shall default to REVIEW with an explanatory placeholder rather than being omitted.
FR-4.4	For the “Payment Terms” item, the system shall deterministically set the status by comparing the RFP's stated payment terms (in days) against the Company Profile's acceptable_payment_terms_days: at or under the threshold → GO; over → NO-GO.

FR-4.5	For the “Insurance Requirements” item, the system shall deterministically set the status by comparing the RFP's stated insurance requirement against the Company Profile's max_insurance_available_usd: at or under → GO; over → NO-GO.
FR-4.6	If either FR-4.4 or FR-4.5 results in NO-GO, the overall verdict (FR-3.1) shall be adjusted: an insurance breach forces the overall verdict to NO-GO; a payment-terms breach (with no insurance breach) forces at least CONDITIONAL.

FR-5: Department & Overall Scoring

ID	Requirement
FR-5.1	The system shall compute a 0–100% compliance score for each of the four departments and an overall score, derived directly from checklist item statuses (GO=100, REVIEW=50, NO-GO=0, averaged) — not as a separate AI judgment.
FR-5.2	The system shall assign a recommendation label per department and overall: “Proceed” if all items are GO; “Review Needed” if any item is REVIEW (and none are NO-GO); “High Risk” if any item is NO-GO.

FR-6: Results Presentation & Export

ID	Requirement
FR-6.1	The system shall present results in a tabbed interface: Deliverables, Evaluation Criteria, Compliance Checklist, Dates & Budget, Strengths & Risks.
FR-6.2	The Compliance Checklist tab shall display the department scorecard (FR-5) above the detailed per-item results.
FR-6.3	The system shall allow the completed analysis to be exported as a Markdown (.md) file.
FR-6.4	The system shall allow the completed analysis to be exported as a styled PDF report, including the verdict, scorecard, deliverables, criteria, dates/budget, strengths/risks, and full compliance checklist.

FR-7: Error Handling & Resilience

ID	Requirement
FR-7.1	The system shall automatically retry transient API errors (HTTP 429 rate limits, 500/503 server errors) with exponential backoff, up to a configured maximum number of attempts.
FR-7.2	The system shall distinguish a daily quota exhaustion error from a transient rate limit and fail immediately with a clear message in that case, rather than retrying uselessly.
FR-7.3	If any single department's compliance call fails after retries, the system shall continue processing the remaining departments, mark the failed department's items as REVIEW, and surface a visible warning naming the failed department, rather than aborting the entire analysis.

3.2 External Interface Requirements

3.2.1 User Interface

A Streamlit web application: a sidebar for API key entry and Company Profile editing, a main panel for file upload and the Analyze action, and a tabbed results view as described in FR-6.

3.2.2 Software Interfaces

- Google Gemini API (via the google-genai SDK) — model gemini-2.5-flash, using structured JSON output (response_schema) for all five calls per analysis.
- pdfplumber — PDF text extraction.
- ReportLab — PDF report generation.

3.2.3 Communications Interfaces

HTTPS calls to Google's generativelanguage.googleapis.com endpoint via the google-genai SDK. No other network communication is performed by the system in its current scope.

3.3 Non-Functional Requirements

NFR-1: Performance

- A single analysis run performs 5 sequential API calls (1 qualification + 4 department checklists); typical completion time is dominated by Gemini response latency, not local processing.

- PDF text extraction shall complete in under 5 seconds for documents up to approximately 25 pages.

NFR-2: Reliability

- The system shall not lose previously extracted RFP text or Company Profile edits due to unrelated UI interactions (e.g. switching tabs), per the session-state design described in Section 4.
- A failure in one part of the compliance checklist (FR-7.3) shall not cause loss of the qualification assessment (FR-3) or the other three departments' results.

NFR-3: Usability

- No coding knowledge shall be required to operate the tool: upload, edit profile fields via standard form widgets, click Analyze, read tabbed results.

NFR-4: Security

- The Gemini API key shall be accepted via a password-masked input field or a local .env file, and shall not be persisted to disk by the application itself beyond the current session.

NFR-5: Maintainability

- The checklist (35 items) shall be defined in a single, isolated module (checklist_items.py) so it can be updated without touching AI prompt logic, scoring logic, or UI code.
- The two deterministic threshold rules (FR-4.4, FR-4.5) shall be isolated in their own module (decision_rules.py), separate from AI-calling code, so policy changes (e.g. a different insurance ceiling) don't require touching prompt engineering.

4. System Architecture

The analysis pipeline, per run, executes the following calls in sequence:

9. Core Analysis call — requests verdict, deliverables, evaluation criteria, key dates/budget, strengths, and risks in one schema-constrained response.
10. Financial Compliance call — requests GO/NO-GO/REVIEW + reason + evidence + page reference for exactly 6 items.
11. Legal Compliance call — same, for exactly 13 items.
12. Operations Compliance call — same, for exactly 11 items.
13. Technical Compliance call — same, for exactly 5 items.

Each compliance call uses a dynamically generated Pydantic schema with an exact min/max array length matching that department's item count (Section 4.1 explains why this is 4 calls rather than 1).

4.1 Design Rationale: Why the Checklist Is Split Into 4 Calls

An earlier version of this system requested all 35 checklist items in a single call, constrained by a schema requiring exactly 35 array entries. In testing against a long, complex real-world RFP, this produced two distinct failures at different times:

- On some documents, the model's response validated as JSON but the compliance array was empty or mostly empty (the model prioritized other sections of the shared response and did not complete the large repetitive section).
- After adding a strict schema-level minimum/maximum length of 35 to force completeness, Gemini's API instead rejected the request outright with a 400 error: "The specified schema produces a constraint that has too many states for serving" — a documented limitation of Gemini's controlled-generation feature when an exact-length array of complex nested objects (each containing an enum field) grows too large.

The resolution was to split the checklist into four department-scoped calls, each with its own small exact-length schema (6, 13, 11, and 5 items respectively). This keeps each call's constraint complexity within Gemini's serving limits while preserving the guarantee that a valid response for a given department contains exactly that department's items — and isolates failures to a single department (FR-7.3) instead of risking the whole checklist.

4.2 Merge & Override Logic

After all five calls complete (or fail, per FR-7.3), results are merged back onto the fixed 35-item list by exact item name — not by array position — so the final report always contains all 35 items in a fixed, predictable order regardless of how the model orders or labels its response. The two deterministic rules (FR-4.4, FR-4.5) are then applied on top, overwriting whatever the model returned for those two specific items, and department/overall scores (FR-5) are computed from the final merged list.

5. Data Requirements

5.1 Company Profile (default values)

Field	Default value
company_name	SPS
services	Identity and Access Management (IAM), cybersecurity solutions, SOC/SIEM monitoring
years_experience	8
max_insurance_available_usd	\$5,000,000
acceptable_payment_terms_days	30 (NET30)
certifications	NIST 800-53 aligned, SOC 2, E-Verify enrolled
annual_revenue_usd	\$12,000,000
can_provide_audited_financials	True
registered_states	Home state only (register elsewhere on a per-bid basis)

5.2 The 35-Item Compliance Checklist

Ported directly from SPS's Financial/Legal/Operations/Technical checklist source document; treated as fixed, literal ground truth (Section 2.4).

Financial / Accounting (6 items)

- Payment Terms
- Financial Stability Requirements
- Unaudited Financial Statements
- Insurance Requirements
- Profitability Analysis
- Bid Bond

Legal (13 items)

- Relevant Experience
- Registration Requirement
- Financial Statement of Previous Year
- Qualified Personnel
- Technical Knowhow
- Expected Revenue Generation
- Period of Implementation
- Insurance Coverage
- Compliance of Law
- Compliance Requirements (Data Protection)
- State Registration
- E-Verify
- Contractual Obligations

Operations (11 items)

- Insurance Requirement Form
- Information Form (Tax ID, Owner Name, % Ownership)
- Small Business Certification
- MBE Certification
- Workers Comp Insurance
- Business with Iran
- Submission Deadlines
- Document Compliance
- Signatory Authority
- Vendor Registration
- Checklist of Required Documents

Technical (5 items)

- Scope of Services/Products Alignment
- Technical Requirements
- Compliance with Industry Standards
- Security Considerations
- Integration Needs

6. Known Limitations & Out-of-Scope Items

- No native support for scanned/image-only PDFs (no OCR).
- The 35-item checklist is fixed in code; adapting it to a different set of internal policies requires a code change.
- Gemini's free tier rate/quota limits apply; heavy concurrent use requires a billed Google Cloud project.
- Stages 4–7 of the parent Proposal Capture System assignment — Knowledge Retrieval, Proposal Generation, Review & Refinement, and Orchestration — are not implemented and are explicitly out of scope for this document.
- The Proposal Capture Design Document and Final Presentation deliverables of the parent assignment are separate written/spoken deliverables, not covered by this SRS.